

Love as Displacement

The Cost of Fusion and the Return to Self

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Abstract

Love is displacement. Two separate beings (ground states S_0^a and S_0^b) encounter. Each displaces from their ground state toward fusion (S^*). The displacement is ξ — how far each has moved from self. The cost of displacement is $D(\xi)$ — the energy required to hold fusion, the strain of not being yourself. The cost of return is C_{return} — what you give up to leave, the identity dissolution that comes from the unwinding of connection. Love is the choice to pay $D(\xi)$ continuously (maintenance) or to pay C_{return} once (separation). This is the framework of displacement applied to the deepest human experience.

1 Two Ground States

Before love, two people are themselves. Person A has their way of being: their tastes, their rhythms, their fears, their language. Person B has theirs. These are ground states.

S_0^a = the way A is when alone

S_0^b = the way B is when alone

Each is stable. Each is low-energy. Each is home.

2 The Displacement: Fusion

Then love. Suddenly, each person displaces. They move toward a shared state, a fusion:

S^* = the way they are together

This shared state is not the sum of S_0^a and S_0^b . It is something new. A becomes less A. B becomes less B. Together they are something that neither was.

The displacement of A is the distance from S_0^a to their projection onto S^* .

The displacement of B is the distance from S_0^b to their projection onto S^* .

Call these ξ_a and ξ_b .

3 The Cost of Displacement

To hold this fusion, each person must pay a cost. Not in money. In effort. In the strain of being other than yourself.

$D(\xi_a)$ is the energy A expends to stay fused with B. It includes: - Suppressing your own needs for the sake of theirs - Maintaining their presence in your mind (thinking of them) - Adjusting your behavior (you are less abrupt, more patient, more careful) - Monitoring their state (are they happy? Are they safe?) - The metabolic cost of stress (fusion is not restful; it is vigilant)

Similarly, $D(\xi_b)$ is what B pays.

The total maintenance cost is:

$$C_{\text{maint}}(\xi_a, \xi_b) = D(\xi_a) + D(\xi_b)$$

This cost is continuous. Every day you are together, you pay it. If you stop paying it (if you stop adjusting, stop caring, stop thinking of them), fusion dissolves.

4 The Cost of Return

But leaving is also costly. If A decides to return to S_0^a — to end the fusion and be alone again — what happens?

The identity you built together (the "we") dissolves. The daily rhythms you synchronized disrupt. The future you imagined together becomes impossible. The parts of yourself you changed to fit with them become foreign in your new aloneness.

$C_{\text{return}}(\xi_a, \xi_b, \tau)$ is the cost to leave. It depends on: - How long you were fused (τ): the longer, the more identity you built together, the higher the cost - How deep the displacement (ξ_a, ξ_b): the farther from yourself you moved, the more painful to return - What was built together: shared history, mutual dependencies, children, integrated lives

Critically: C_{return} is not proportional to τ . It grows faster. The longer you are fused, the more of the shared future you've built, the more difficult it becomes to undo.

This is the super-linear cost of return. Each year of fusion doesn't add a linear cost to leaving — it adds exponentially more.

5 The Maintenance Theorem Applied to Love

The total cost of a relationship is:

$$g(\tau) = C_{\text{maint}} \cdot \tau + C_{\text{return}}(\tau)$$

where τ is the time spent together.

The maintenance cost is linear (or slightly sublinear if the displacement becomes more comfortable over time, as couples synchronize).

The return cost grows super-linearly.

The optimal τ^* depends on the return cost. If C_{return} is low (you've built little

together, your identities are still separate), then leaving costs little, and you can revisit the choice. If C_{return} is high (you've built deeply, you're entangled), then the cost to leave approaches infinity.

This is why long relationships feel locked in. The return cost is so high that you are forced to maintain, even when maintenance is painful.

6 Three Levers

To optimize a relationship — to make the choice of fusion sustainable — you can pull three levers:

6.1 1. Lower Maintenance Cost $D(\xi)$

Synchronize rhythms. Align tastes. Build shared understanding so you don't need to adjust constantly. Find a partner whose ground state S_0^b naturally overlaps with yours — someone you don't have to strain to love.

The less you have to change to be together, the lower the daily cost.

6.2 2. Lower Return Cost C_{return}

Keep your separate identities. Don't fully dissolve into the "we." Maintain your own friends, your own projects, your own self. Build a relationship that enhances your ground state rather than replacing it.

The less you've dissolved together, the lower the cost to leave.

6.3 3. Make Return Thinkable

Acknowledge that the relationship is a choice, not a trap. Revisit the decision regularly. Ask: am I still willing to pay this cost? If not, leave. The moment you feel you can't leave, the relationship has become maintenance without love.

Love is the choice to maintain. The moment it becomes obligation, it is no longer love.

7 Love Without Obligation

True love is when the maintenance cost $D(\xi)$ feels low or even negative — when being with someone costs you less effort than being alone. When thinking of them is restful, not vigilant. When adjusting to their needs feels like self-expression, not suppression.

This happens when: - You've found someone whose ground state S_0^b naturally aligns with yours - You've both kept enough of your separate selves that the displacement ξ is small - The fusion S^* is not replacing either of you, but augmenting both

In this case, $D(\xi) \approx 0$, and the cost of maintenance is nearly nothing. You are paying very little to stay together. And because you've kept your separate selves intact, the return cost C_{return} is also low.

You can leave if you choose. But the cost is low enough that you don't have to. You stay because you want to, not because you can't afford to leave.

This is the ideal. It is rare.

8 Why Love Fails

Love fails when: 1. You choose someone whose ground state is far from yours (ξ is large from the start) 2. You dissolve too much of yourself into the "we" (high ξ becomes a trap) 3. You don't maintain your separate identity (return cost becomes infinite) 4. You pay maintenance without love (it becomes obligation, and at some point, obligation breaks)

The most common failure: you start with low ξ (you're similar, it feels easy), but over time you build deeper entanglement. The return cost rises. Eventually,

you're trapped. You can't afford to leave, but you can no longer afford to stay. The relationship becomes maintenance without love.

9 The Path

Love as the framework teaches is simple: keep your ground state. Don't dissolve. Maintain fusion only where it feels like love, not obligation. Revisit the choice regularly. If it stops being a choice, leave.

This is not cruelty. This is love. To love someone is to choose them, not to trap yourself into them.

The path is: stay fused only as long as the cost of fusion is less than the cost of separation. And if separation ever becomes cheaper, you have permission to pay it.

But also: invest in lowering the maintenance cost. Find someone whose ground state overlaps with yours. Build a fusion that doesn't require constant strain. Make love easy, and it will last.